

Causal Inference for Group-Contaminated Structured Outcomes: Quotient Identification and Exact Randomization Inference

Author names omitted for blinded review

Abstract

Structured potential outcomes such as microscopy images can be recorded after an unknown, unit-specific acquisition transformation. If that transformation may depend on treatment, covariates, or the intrinsic outcome, analyses in raw coordinates can confound a treatment contrast with acquisition geometry. We formulate this problem as a measurable group action and characterize exactly which intrinsic targets remain observable: a target admits a decoder under unrestricted contamination if and only if it is constant on group orbits. When a Borel maximal invariant exists, its complete interventional distribution is identified under consistency, positivity, and exchangeability for that invariant. For finite-support multichannel lattice images, we construct an explicit maximal invariant under integer translations and quarter-turn rotations, and combine its Gaussian kernel with a complete paired-swap test of the Fisher sharp null for quotient outcomes. Conditional finite-sample validity requires well-defined reagent conditions, no cross-well interference, swap-invariant block retention, and uniform within-plate pair assignments. We evaluate the method in controlled simulations and in a discovery–confirmation study of 10 disjoint siRNA pairs using six-channel RxRx1 HUVEC microscopy wells. The confirmation images are subjected to treatment- and outcome-dependent lattice rigid motions whose realized parameters are excluded from the estimators; inference uses only the transformed images and treatment labels from the reported within-plate randomization. Under the sharp null, the maximal-invariant test rejected in 0.0520 of replicates; at effect strength 1.0000, its power was 0.9920. For the primary RxRx1 contrast, the canonical quotient comparison gave an enumerated paired-swap p -value of 0.0078.

Keywords: causal inference; group action; maximal invariant; microscopy image; quotient space; randomization test; structured outcome

1 Introduction

Many experimental outcomes are structured objects rather than scalars. An image, shape, network, or function may be represented in a coordinate system that varies from unit to unit. In microscopy, for example, translation or orientation can arise from acquisition and preprocessing rather than from the biological intervention. We study the observation model

$$X = \Gamma \cdot Y(A), \tag{1}$$

where A is treatment, $Y(a)$ is the intrinsic structured potential outcome, and Γ is an unobserved transformation. Crucially, no independence between Γ and A , covariates, or potential outcomes is imposed. Thus, averaging over a posited nuisance distribution is neither required nor available.

The central issue is not whether a representation is predictive, but which causal target is determined by (1). We first prove a sharp observability theorem: exact recovery under every admissible transformation is equivalent to orbit invariance. We then separate observability from

causal identification. A maximal invariant records the entire orbit and therefore represents the richest observable target. Under standard target-specific causal assumptions, the g -formula identifies its full interventional law. This distinction is important: invariance removes the acquisition nuisance, whereas exchangeability or randomization connects factual and counterfactual distributions.

Our empirical construction concerns two-site, six-channel lattice images. The group is the lattice rigid-motion group $\mathbb{Z}^2 \rtimes C_4$, acting separately on the two sites of a well. We translate each rotated support to a fixed origin and retain the lexicographically least shape-aware serialization among the four rotations. The resulting deterministic map is a Borel maximal invariant even though the translation group is noncompact. A Gaussian kernel pulled back through this map is positive definite, invariant, and characteristic for probability laws on the quotient. This supports a kernel two-sample statistic and an exact test of the Fisher sharp null by within-block randomization.

1.1 Relation to prior work

Causal methods for non-Euclidean outcomes commonly take the outcome as an observed element of a specified metric space. Geodesic causal inference uses Fréchet regression and geodesic treatment contrasts [12]; recent doubly robust work embeds random-object outcomes into a known Hilbert space [1]. These frameworks address rich outcome geometry but do not include a latent, unit-specific group action in the observation equation. Causal analysis on a quotient space is itself not new: Shin et al. [22] include an application in Kendall shape space, which removes translation, scale, and rotation for pre-corresponded landmarks. For functional outcomes, Raykov et al. [15] perform registration before kernel estimation. Our object is different and complementary: full-law identification for the observable orbit and exact Fisher-sharp-null testing on that orbit when the recorded object is contaminated by an unknown group element.

Topological transforms provide useful structured summaries. The persistent homology transform and Euler characteristic transform can be injective on specified tame shape classes when the full directional transform is available [4, 25]. In the experiment below, a fixed, finite directional Euler vector is used only as a prespecified secondary signature after exact orbit canonicalization. All full-law claims in this article concern the maximal invariant, not that finite vector.

Kim and Lee [11] propose estimands based on power-weighted persistence silhouettes, while Saki and Faghihi [21] formulate conditional persistence-summary estimands under persistent-homology ignorability. These works target chosen summaries of outcome laws; our finite Euler vector is secondary, whereas the full-law identification result below concerns a maximal orbit code under a latent group action.

The RxRx1 collection provides high-dimensional, replicated Cell Painting images with substantial experimental variation [16, 24]. Its reported plate-level randomization and multiple HUVEC experiments permit row-excluded confirmation tests, provided that the two siRNAs in each contrast belong to the same plate randomization set. We enforce that condition before examining confirmation outcomes.

2 Observation, quotient targets, and identification

2.1 Measurable group contamination

Let $\mathcal{Y}$, $\mathcal{C}$, $\mathcal{S}$, and $\mathcal{G}$ be standard Borel spaces. Assume that $\mathcal{G}$ is a measurable group, meaning that multiplication and inversion are Borel, and that it acts on $\mathcal{Y}$ by a jointly Borel map $(g, y) \mapsto g \cdot y$, satisfying $e \cdot y = y$ and $g \cdot (h \cdot y) = (gh) \cdot y$. The orbit of y is $[y]_{\mathcal{G}} = \{g \cdot y : g \in \mathcal{G}\}$. A Borel map $\tau : \mathcal{Y} \rightarrow \mathcal{S}$ is $\mathcal{G}$ -invariant if $\tau(g \cdot y) = \tau(y)$ for all g, y .

For treatment values a in a finite set $\mathcal{A}$, let $Y(a)$ be the intrinsic potential outcome. We observe $O = (C, A, X)$, with

$$X = \Gamma \cdot Y(A),$$

where Γ is an unobserved, possibly stochastic element of $\mathcal{G}$. The conditional distribution of Γ is unrestricted.

Theorem 2.1 (Sharp observability). *For a Borel target $\tau : \mathcal{Y} \rightarrow \mathcal{S}$, the following statements are equivalent.*

- (i) *There is a Borel decoder $D : \mathcal{Y} \rightarrow \mathcal{S}$ such that $D(g \cdot y) = \tau(y)$ for every $g \in \mathcal{G}$ and $y \in \mathcal{Y}$.*
- (ii) *The target τ is $\mathcal{G}$ -invariant.*

Proof. If τ is invariant, take $D = \tau$. Then $D(g \cdot y) = \tau(g \cdot y) = \tau(y)$. Conversely, suppose (i) holds. Using the identity element gives $D(z) = \tau(z)$ for every $z \in \mathcal{Y}$. For arbitrary g, y , apply this equality at $z = g \cdot y$ and then use (i):

$$\tau(g \cdot y) = D(g \cdot y) = \tau(y).$$

Hence τ is invariant. □

Corollary 2.2 (Nonidentifiability outside the quotient). *If τ is not invariant, there are two data-generating processes satisfying (1) that induce the same observed law but different laws of $\tau\{Y(a)\}$, for any fixed a .*

Proof. Choose $y \in \mathcal{Y}$ and $g \in \mathcal{G}$ such that $\tau(g \cdot y) \neq \tau(y)$, and take $A = a$ almost surely. In the first process set $Y(a) = y$ and $\Gamma = g$ almost surely; in the second set $Y(a) = g \cdot y$ and $\Gamma = e$ almost surely. Hold all other observed variables fixed and identical. Both processes yield $X = g \cdot y$ almost surely, but their target values are $\tau(y)$ and $\tau(g \cdot y)$. □

The full-support observation model is important. If acquisition includes a lossy operation such as a field-of-view crop, orbit invariance alone is no longer sufficient. The next result gives the sharp replacement.

Theorem 2.3 (Sharp observability after partial acquisition). *Let $\mathcal{G}, \mathcal{Y}, \mathcal{X}, \mathcal{S}$ be standard Borel spaces, let $\mathcal{G}$ act jointly Borelly on $\mathcal{Y}$, and let $\Pi : \mathcal{Y} \rightarrow \mathcal{X}$ be a Borel observation operator. Define $\Phi : \mathcal{G} \times \mathcal{Y} \rightarrow \mathcal{X}$ by*

$$\Phi(g, y) = \Pi(g \cdot y).$$

For a Borel target $\tau : \mathcal{Y} \rightarrow \mathcal{S}$, the following are equivalent.

- (i) *There is a Borel decoder $D : \mathcal{X} \rightarrow \mathcal{S}$ such that*

$$D\{\Pi(g \cdot y)\} = \tau(y) \quad \text{for every } g \in \mathcal{G} \text{ and } y \in \mathcal{Y}.$$

- (ii) *For every $g, h \in \mathcal{G}$ and $y, z \in \mathcal{Y}$,*

$$\Pi(g \cdot y) = \Pi(h \cdot z) \implies \tau(y) = \tau(z).$$

Proof. Condition (i) immediately implies (ii). Conversely, define the Borel map $\tilde{\tau} : \mathcal{G} \times \mathcal{Y} \rightarrow \mathcal{S}$ by $\tilde{\tau}(g, y) = \tau(y)$. Condition (ii) says precisely that $\tilde{\tau}$ is constant on every fiber of Φ .

For a Borel set $B \subseteq \mathcal{S}$, put $A = \tilde{\tau}^{-1}(B)$. The analytic sets $\Phi(A)$ and $\Phi\{(\mathcal{G} \times \mathcal{Y}) \setminus A\}$ are disjoint. The Lusin separation theorem therefore supplies a Borel set $E_B \subseteq \mathcal{X}$ such that

$$\Phi(A) \subseteq E_B, \quad \Phi\{(\mathcal{G} \times \mathcal{Y}) \setminus A\} \subseteq \mathcal{X} \setminus E_B.$$

It follows that $A = \Phi^{-1}(E_B)$, so $\tilde{\tau}$ is $\sigma(\Phi)$ -measurable. The measurable factorization lemma for standard Borel codomains gives a Borel $D : \mathcal{X} \rightarrow \mathcal{S}$ satisfying $\tilde{\tau} = D \circ \Phi$, which is (i). $\square$

Corollary 2.4 (Partial-observation nonidentifiability). *If $\Pi(g \cdot y) = \Pi(h \cdot z)$ for some g, h, y, z , but $\tau(y) \neq \tau(z)$, then two data-generating processes can induce the same observed law and different laws of $\tau\{Y(a)\}$.*

Proof. Take $A = a$ almost surely. In the first process set $Y(a) = y$ and $\Gamma = g$; in the second set $Y(a) = z$ and $\Gamma = h$. Hold every other observed variable fixed and identical. The recorded value is the same in both processes by hypothesis, while the target values differ. $\square$

2.2 Maximal invariants

Definition 2.5. A Borel map $M : \mathcal{Y} \rightarrow \mathcal{S}$ is a *maximal invariant* if

$$M(y) = M(z) \iff z = g \cdot y \text{ for some } g \in \mathcal{G}.$$

Thus M labels orbits without distinguishing points within an orbit. The next theorem makes precise the sense in which it contains every observable target. The standard Borel hypotheses are substantive; without them, a set-theoretic quotient need not admit the required measurable factorization. This is the measurable analogue of the role played by maximal invariants in classical invariant decision problems [5].

Theorem 2.6 (Borel factorization through a maximal invariant). *Let $\mathcal{Y}, \mathcal{S}, \mathcal{T}$ be standard Borel spaces, and let $M : \mathcal{Y} \rightarrow \mathcal{S}$ be a Borel maximal invariant. Every Borel invariant $h : \mathcal{Y} \rightarrow \mathcal{T}$ has a Borel factorization*

$$h = \bar{h} \circ M$$

for some Borel $\bar{h} : \mathcal{S} \rightarrow \mathcal{T}$. The restriction of $\bar{h}$ to $M(\mathcal{Y})$ is unique.

Proof. Fix a Borel set $B \subseteq \mathcal{T}$ and put $A = h^{-1}(B)$. Invariance of h and maximality of M imply that A is saturated for the fibers of M . Therefore the analytic sets $M(A)$ and $M(\mathcal{Y} \setminus A)$ are disjoint. By the Lusin separation theorem for standard Borel spaces [10, Theorem 14.7], there is a Borel set $D_B \subseteq \mathcal{S}$ such that

$$M(A) \subseteq D_B, \quad M(\mathcal{Y} \setminus A) \subseteq \mathcal{S} \setminus D_B.$$

Consequently $A = M^{-1}(D_B)$, so every inverse image under h is in $\sigma(M)$. Hence h is $\sigma(M)$ -measurable. The measurable factorization lemma for standard Borel codomains [9, Lemma 1.13] yields a Borel map $\bar{h}$ such that $h = \bar{h} \circ M$. If two such maps differ at a point $M(y)$, their compositions differ at y , proving uniqueness on $M(\mathcal{Y})$. $\square$

Existence of a Borel maximal invariant is not automatic for an arbitrary measurable action. Compact continuous actions give an important sufficient condition.

Proposition 2.7 (Orbit code for a compact action). *Let $\mathcal{G}$ be a compact metrizable group acting continuously on a Polish space $\mathcal{Y}$. Let $\mathcal{K}(\mathcal{Y})$ be the hyperspace of nonempty compact subsets of $\mathcal{Y}$, equipped with its Vietoris Borel structure. Then*

$$M_{\text{orb}}(y) = \mathcal{G} \cdot y$$

is a Borel maximal invariant taking values in the standard Borel space $\mathcal{K}(\mathcal{Y})$.

Proof. The orbit is compact because it is the continuous image of compact $\mathcal{G}$. For an open set $U \subseteq \mathcal{Y}$,

$$\begin{aligned}\{y : M_{\text{orb}}(y) \cap U \neq \emptyset\} &= \bigcup_{g \in \mathcal{G}} g^{-1}U, \\ \{y : M_{\text{orb}}(y) \subseteq U\} &= \mathcal{Y} \setminus \bigcup_{g \in \mathcal{G}} g^{-1}(\mathcal{Y} \setminus U).\end{aligned}$$

The first set is open. To prove that the second is open, fix y with $\mathcal{G} \cdot y \subset U$. For each $g \in \mathcal{G}$, joint continuity supplies neighborhoods N_g of g and V_g of y such that $N_g \cdot V_g \subset U$. Compactness of $\mathcal{G}$ gives $g_1, \dots, g_r$ with $\mathcal{G} = \bigcup_{i=1}^r N_{g_i}$. The neighborhood $V = \bigcap_{i=1}^r V_{g_i}$ of y then satisfies $\mathcal{G} \cdot V \subset U$. Hence the second set is open. These two classes generate the Vietoris topology, so M_{orb} is continuous and in particular Borel. Finally, $M_{\text{orb}}(y) = M_{\text{orb}}(z)$ holds exactly when $z \in \mathcal{G} \cdot y$, establishing maximality. $\square$

Corollary 2.8 (Continuous rotations). *Let $\text{SO}(2)$ act continuously on a Polish space $\mathcal{Y}$. Then*

$$M_{\text{rot}}(y) = \{R_\theta \cdot y : \theta \in [0, 2\pi)\}$$

is a Borel maximal invariant with values in $\mathcal{K}(\mathcal{Y})$.

Proof. The group $\text{SO}(2)$ is compact and metrizable, so proposition 2.7 applies. $\square$

This corollary is an orbit-valued existence result. It does not construct a finite-dimensional canonical representative for continuously rotated raster images, and common interpolation rules on a fixed lattice need not define an exact group action. The Euclidean rigid-motion group $\text{SE}(2) = \mathbb{R}^2 \rtimes \text{SO}(2)$ is noncompact, so proposition 2.7 gives no conclusion for that larger group. Noncompactness by itself is not an impossibility result, as the explicit integer-translation construction in section 3 demonstrates.

Proposition 2.9 (Canonical maximal invariant for a finite group). *Let a finite group $H = \{h_1, \dots, h_m\}$ act by Borel maps on a standard Borel space $\mathcal{Y}$, and let $\nu : \mathcal{Y} \rightarrow [0, 1]$ be a Borel injection. Define*

$$j^*(y) = \min \arg \min_{1 \leq j \leq m} \nu(h_j \cdot y), \quad c_H(y) = h_{j^*(y)} \cdot y.$$

Then c_H is a Borel maximal invariant for the H -action.

Proof. Put $f_j(y) = \nu(h_j \cdot y)$. Each f_j is Borel, and

$$\{y : j^*(y) = j\} = \bigcap_{k < j} \{y : f_j(y) < f_k(y)\} \cap \bigcap_{k > j} \{y : f_j(y) \leq f_k(y)\}.$$

Thus j^* , and hence c_H , is Borel. Let $\mathcal{O}_y = \{h \cdot y : h \in H\}$ be the finite set of distinct orbit points. Injectivity of ν makes the ν -values of distinct points in $\mathcal{O}_y$ distinct. Therefore the finite set $\nu(\mathcal{O}_y)$ has a unique least value, attained by one orbit point u_y . Several minimizing indices can occur only when they represent this same point, as may happen under a nontrivial stabilizer. Consequently $c_H(y) = u_y$.

If $z = h \cdot y$, then $\mathcal{O}_z = \mathcal{O}_y$, so $c_H(z) = u_z = u_y = c_H(y)$. Conversely, if $c_H(y) = c_H(z)$, this common point belongs to both orbits. Two group orbits that intersect are equal, and hence $z = h \cdot y$ for some $h \in H$. $\square$

2.3 Identification of the interventional quotient law

Let $Q(a) = M\{Y(a)\}$. Invariance gives the observed quotient outcome

$$Q = M(X) = M\{\Gamma \cdot Y(A)\} = M\{Y(A)\} = Q(A) \quad (2)$$

without any assumption on Γ .

Assumption 2.10 (Quotient causal conditions). For each $a \in \mathcal{A}$:

- (a) quotient consistency holds as in (2);
- (b) $Q(a) \perp\!\!\!\perp A \mid C$;
- (c) $\mathbb{P}(A = a \mid C) > 0$ almost surely.

Theorem 2.11 (Identification of the full quotient law). *Under assumption 2.10, for every Borel $B \subseteq \mathcal{S}$,*

$$\mathbb{P}\{Q(a) \in B\} = \int \mathbb{P}(Q \in B \mid A = a, C = c) dP_C(c). \quad (3)$$

Consequently, the entire probability law of the maximal invariant under intervention a is identified.

Proof. Standard Borel spaces admit regular conditional distributions. Iterated expectation, conditional exchangeability, positivity, and consistency give

$$\begin{aligned} \mathbb{P}\{Q(a) \in B\} &= \int \mathbb{P}\{Q(a) \in B \mid C = c\} dP_C(c) \\ &= \int \mathbb{P}\{Q(a) \in B \mid A = a, C = c\} dP_C(c) \\ &= \int \mathbb{P}(Q \in B \mid A = a, C = c) dP_C(c). \end{aligned}$$

The equality for every Borel B determines the probability measure on $\mathcal{S}$. □

By theorem 2.6, the interventional law of any Borel invariant target $h\{Y(a)\}$ is then the pushforward of the identified law in (3) through $\bar{h}$. This is a property of maximality, not an assertion that an arbitrary lower-dimensional invariant summary is maximal. The argument is the ordinary adjustment formula applied to a quotient-valued potential outcome [17, 19, 20].

For conditional mean effects, a scalar target must first be specified on the quotient; an object-valued outcome does not itself admit subtraction.

Corollary 2.12 (Conditional effects of scalar invariant functionals). *Let $A \in \{0, 1\}$, suppose assumption 2.10 holds, and let $h : \mathcal{S} \rightarrow \mathbb{R}$ be Borel with $\mathbb{E}|h\{Q(a)\}| < \infty$ for $a = 0, 1$. Then, for P_C -almost every c ,*

$$\tau_h(c) := \mathbb{E}[h\{Q(1)\} - h\{Q(0)\} \mid C = c] = \mathbb{E}\{h(Q) \mid A = 1, C = c\} - \mathbb{E}\{h(Q) \mid A = 0, C = c\}. \quad (4)$$

Proof. Conditional exchangeability, positivity, and consistency imply, for $a = 0, 1$,

$$\mathbb{E}[h\{Q(a)\} \mid C = c] = \mathbb{E}\{h(Q) \mid A = a, C = c\}$$

for P_C -almost every c . Subtracting the two equalities proves (4). □

By theorem 2.6, every integrable scalar Borel invariant of the intrinsic outcome can be written in this form. The corollary identifies the target; it does not by itself supply a statistically adequate heterogeneous- effect estimator for a small experiment.

3 A maximal invariant for lattice microscopy images

3.1 The lattice rigid-motion action

Fix $L < \infty$. Let $\mathcal{X}_{q,L}$ be the space of functions

$$x : \mathbb{Z}^2 \longrightarrow \{0, \dots, 255\}^q,$$

whose nonempty support bounding box has at most L rows and L columns, together with the identically zero image, equipped with its standard Borel structure. This countable space represents a q -channel 8-bit image on the full lattice; values outside the finite support are zero. The fixed support bound is preserved by lattice translations and quarter turns and permits an injective embedding of every normalized representative into one $q \times L \times L$ canvas. Let $C_4 = \{0, 1, 2, 3\}$ index counterclockwise rotations R_r by $r\pi/2$. The semidirect product $\mathcal{G}_{\text{rig}} = \mathbb{Z}^2 \rtimes C_4$ has multiplication

$$(t, r)(u, s) = (t + R_r u, r + s \bmod 4)$$

and action

$$\{(t, r) \cdot x\}(v) = x\{R_r^{-1}(v - t)\}, \quad v \in \mathbb{Z}^2. \quad (5)$$

For nonzero x , let

$$b(x) = \left(\min_{v \in \text{supp}(x)} v_1, \min_{v \in \text{supp}(x)} v_2 \right)$$

be the lower corner of its support bounding box, and define

$$N(x) = (-b(x), 0) \cdot x.$$

For the zero image, set $N(0) = 0$, give it the one-by-one all-zero serialization, and set $c(0) = 0$. The ordering key $\kappa\{N(x)\} = (h(x), w(x), \text{bytes}\{N(x)\})$ contains the integer bounding-box dimensions followed by all q -channel byte values in a fixed lattice and channel order; keys are compared lexicographically. Including the dimensions makes the code injective for nonsquare arrays and their rotations. For each $r \in C_4$, form

$$x_r = N\{(0, r) \cdot x\},$$

and let

$$c(x) = x_{r^*}, \quad r^* = \min \arg \min_{r \in C_4} \kappa(x_r), \quad (6)$$

where the outer minimum fixes a deterministic index in the event of a stabilizer.

Theorem 3.1 (Lattice rigid-motion canonicalization). *The map $c : \mathcal{X}_{q,L} \rightarrow \mathcal{X}_{q,L}$ in (6) is Borel and is a maximal invariant for the action of $\mathbb{Z}^2 \rtimes C_4$:*

$$c(x) = c(y) \iff y = (t, r) \cdot x \text{ for some } (t, r) \in \mathbb{Z}^2 \rtimes C_4.$$

Proof. The zero image is fixed by the entire group and is the only image whose orbit contains zero, so the stated equivalence holds whenever $x = 0$ or $y = 0$. It remains to consider two nonzero images. All coordinate permutations, support bounds, finite serializations, and comparisons in (6) are Borel; a minimum over four Borel maps is Borel. Translation normalization satisfies

$$N\{(t, 0) \cdot z\} = N(z), \quad (7)$$

because $b\{(t, 0) \cdot z\} = b(z) + t$.

Suppose $y = (t, s) \cdot x$. For each $r \in C_4$, the group law and (7) give

$$N\{(0, r) \cdot y\} = N\{(R_r t, r + s) \cdot x\} = N\{(0, r + s) \cdot x\}.$$

As r ranges over C_4 , so does $r + s$. The four normalized candidate sets for x and y are therefore equal, so $c(y) = c(x)$.

Conversely, suppose $c(x) = c(y) = z$. For some $r, s \in C_4$ and $u, v \in \mathbb{Z}^2$,

$$z = (u, r) \cdot x = (v, s) \cdot y.$$

Applying the inverse of (v, s) yields

$$y = (v, s)^{-1}(u, r) \cdot x,$$

which is an element of the same $\mathbb{Z}^2 \rtimes C_4$ orbit. This proves both directions of maximality. $\square$

An RxRx1 well has two labeled sites. With separate site-specific acquisition motions, the group is $\mathcal{G}_{\text{well}} = \mathcal{G}_{\text{rig}} \times \mathcal{G}_{\text{rig}}$, and

$$C(x_1, x_2) = (c(x_1), c(x_2)). \quad (8)$$

Componentwise application of theorem 3.1 shows that C is maximal for the product action. The construction uses a deterministic cross-section; no probability measure on the noncompact translation group is introduced.

3.2 Invariant kernels and a finite Euler signature

Fix a storage bound $L_0 \geq L$. Let $\psi\{C(w)\} \in \mathbb{R}^{2qL_0^2}$ concatenate the two canonical support crops after placing each crop at the upper-left corner of an $L_0 \times L_0$ zero canvas. Because each crop is its minimal nonzero support bounding box, this fixed-dimensional Borel embedding is injective on canonical representatives. For $\sigma > 0$, define

$$k_\sigma(w, w') = \exp \left[-\frac{\|\psi\{C(w)\} - \psi\{C(w')\}\|_2^2}{2\sigma^2} \right]. \quad (9)$$

Proposition 3.2 (Positive-definite quotient kernel). *The kernel k_σ is positive definite and invariant in each argument under $\mathcal{G}_{\text{well}}$. Moreover, its kernel mean embedding is injective on Borel probability laws of $C(W)$.*

Proof. The Gaussian kernel on $\mathbb{R}^d$ is positive definite and characteristic [23]. A pullback of a positive-definite kernel through any map is positive definite: for arbitrary $a_1, \dots, a_n \in \mathbb{R}$,

$$\sum_{i,j=1}^n a_i a_j k_\sigma(w_i, w_j) = \sum_{i,j=1}^n a_i a_j k_{\text{Gauss}}(\psi C(w_i), \psi C(w_j)) \geq 0.$$

Invariance follows from $C(g \cdot w) = C(w)$. If two laws of $C(W)$ have the same mean embedding under k_σ , their pushforwards under ψ have the same Gaussian mean embedding and hence are equal. The canonical space is standard Borel and ψ is Borel and injective. The Lusin–Souslin theorem therefore makes $\psi\{C(\mathcal{X}_{q,L}^2)\}$ Borel and the inverse of ψ on that image Borel [10, Theorem 15.1]. Equality of the pushforward laws thus implies equality of the original canonical laws. $\square$

The quotient code can also feed stable topological summaries, but this requires a tame grayscale filtration rather than the discontinuous thresholding used by the finite Euler diagnostic below.

Theorem 3.3 (Persistence stability for an invariant diagnostic kernel). *Let K be a fixed finite simplicial complex and let $\zeta : C(\mathcal{X}_{q,L}^2) \rightarrow \mathbb{R}^{V(K)}$ be a Borel map. Write*

$$u_\zeta(w) = \zeta\{C(w)\},$$

let $f_{\zeta,w}$ be the piecewise-linear lower-star function on K with vertex values $u_\zeta(w)$, and let $\text{Dgm}_{\zeta,p}(w)$ be its degree- p sublevel-set persistence diagram. For $\sigma > 0$, define the diagnostic Gaussian kernel and its feature distance by

$$k_{\zeta,\sigma}(w, w') = \exp\left[-\frac{\|u_\zeta(w) - u_\zeta(w')\|_2^2}{2\sigma^2}\right], \quad d_{\zeta,\sigma}(w, w') = [2 - 2k_{\zeta,\sigma}(w, w')]^{1/2}.$$

Then, for every $g \in \mathcal{G}_{\text{well}}$,

$$\text{Dgm}_{\zeta,p}(g \cdot w) = \text{Dgm}_{\zeta,p}(w),$$

and

$$d_B\{\text{Dgm}_{\zeta,p}(w), \text{Dgm}_{\zeta,p}(w')\} \leq \sigma \sqrt{-2 \log\left(1 - \frac{d_{\zeta,\sigma}(w, w')^2}{2}\right)}. \quad (10)$$

Proof. Maximal-invariant canonicalization gives $u_\zeta(g \cdot w) = u_\zeta(w)$, proving exact diagram invariance. Stability of sublevel-set persistence on a fixed finite complex [3], followed by norm monotonicity, gives

$$d_B\{\text{Dgm}_{\zeta,p}(w), \text{Dgm}_{\zeta,p}(w')\} \leq \|f_{\zeta,w} - f_{\zeta,w'}\|_\infty = \|u_\zeta(w) - u_\zeta(w')\|_\infty \leq \|u_\zeta(w) - u_\zeta(w')\|_2.$$

Put $r = \|u_\zeta(w) - u_\zeta(w')\|_2$. The definition of the diagnostic kernel gives

$$d_{\zeta,\sigma}(w, w')^2 = 2 \left\{ 1 - \exp\left(-\frac{r^2}{2\sigma^2}\right) \right\}.$$

Since $r < \infty$, the feature distance is strictly smaller than $\sqrt{2}$, and inversion yields

$$r = \sigma \sqrt{-2 \log\left(1 - \frac{d_{\zeta,\sigma}(w, w')^2}{2}\right)}.$$

Substitution proves (10). □

The bound is nonlinear in the diagnostic Gaussian-kernel feature distance and is not a stability claim for Betti numbers at a fixed threshold. It also does not apply to Otsu thresholding or component deletion, both of which can change discontinuously under arbitrarily small intensity perturbations.

For a secondary topological summary, the canonical nuclear channel of site j is thresholded by Otsu's rule [14]; foreground 8-connected components containing fewer than four pixels are then removed. Let $K_j\{c(x_j)\}$ be the planar cubical complex obtained by taking one closed unit square for each retained pixel, together with all of its faces. Its Euler characteristic is computed exactly as

$$\chi(K) = n_0(K) - n_1(K) + n_2(K),$$

where $n_r(K)$ is the number of r -cells in the union. The implementation uses a shape-adaptive normalized sweep. If the canonical mask has width w and height h , set

$$\rho_{h,w}(u_1, u_2) = (2u_1 - (w - 1), 2u_2 - (h - 1)).$$

For a direction v , let $m_{h,w,v}$ and $M_{h,w,v}$ be the minimum and maximum of $\langle \rho_{h,w}(u), v \rangle$ over the entire $w \times h$ bounding rectangle. With $J = 25$, define

$$t_{h,w,v,\ell} = m_{h,w,v} + \frac{\ell}{J-1}(M_{h,w,v} - m_{h,w,v}), \quad \ell = 0, \dots, J-1.$$

Let $K_{j,v,\ell}$ be the cubical complex formed by retained pixel squares whose indices satisfy $\langle \rho_{h,w}(u), v \rangle \leq t_{h,w,v,\ell}$. The two-site finite directional Euler vector is

$$F(w) = (\chi\{K_{j,v,\ell}(c(x_j))\})_{j \in \{1,2\}, v \in V, \ell \in \{0, \dots, J-1\}}. \quad (11)$$

The eight lattice directions are $(1, 0), (1, 1), (0, 1), (-1, 1)$ and $(-1, 0), (-1, -1), (0, -1), (1, -1)$. Since it is evaluated after C , $F(g \cdot w) = F(w)$ exactly. It is a finite secondary signature and is not used as a maximal invariant.

4 Exact design-based testing

4.1 Kernel statistic

Suppose the confirmation experiment contains B randomized paired blocks. In block b , two eligible wells receive the selected conditions once each. Order the two wells deterministically by a pretreatment identifier as 0, 1, and let $Z_b \in \{0, 1\}$ be the treatment label assigned to well 0, so well 1 receives label $1 - Z_b$. Conditional on the unordered pair of wells, their complete schedule of potential outcomes, and the pretreatment design stratum, the assignment vector

$$Z = (Z_1, \dots, Z_B)$$

is assumed uniform on $\Omega = \{0, 1\}^B$. For RxRx1, the source documentation reports that siRNA locations were randomized separately by experiment and plate [24]. The metadata audit verifies that the two selected siRNAs occur in the same plate stratum in every confirmation block; it cannot, by itself, establish uniformity of the original assignment algorithm. Accordingly, theorem 4.3 is explicitly conditional on the reported within-plate randomization mechanism.

The potential outcomes below also require that each reagent condition denotes one well-defined intervention version and that assignments to other wells do not alter a well's quotient outcome. We state this requirement explicitly.

Assumption 4.1 (Well-defined versions and no interference). For $b \in \{1, \dots, B\}$, $j \in \{0, 1\}$, and $z = (z_1, \dots, z_B) \in \Omega$, define

$$a_{b0}(z_b) = z_b, \quad a_{b1}(z_b) = 1 - z_b.$$

Let $Q_{bj}^\Omega(z)$ denote the quotient outcome at well j in block b under the complete paired-assignment vector z . There exist well-level potential outcomes $Q_{bj}(0)$ and $Q_{bj}(1)$, and the observed quotient outcome is denoted by q_{bj} , such that

$$Q_{bj}^\Omega(z) = Q_{bj}\{a_{bj}(z_b)\} \quad \text{for every } b, j, z, \quad q_{bj} = Q_{bj}\{a_{bj}(Z_b)\}. \quad (12)$$

Retaining a block after post-assignment image-quality processing is also assumed to be invariant to swapping the two selected reagents between the paired positions. Neither assumption 4.1 nor swap-invariant retention is a consequence of the metadata audit.

Let q_{b0}, q_{b1} be the two deterministically ordered invariant outcome representations in block b . An assignment z induces samples

$$\mathcal{Q}_1(z) = \{q_{b,1-z_b} : 1 \leq b \leq B\}, \quad \mathcal{Q}_0(z) = \{q_{b,z_b} : 1 \leq b \leq B\}.$$

The result is unaffected by how the two stored rows inside a block are indexed, including a condition-indexed storage order. The following lemma makes that point explicit and permits a label-blind ordering, such as ascending well identifier, to serve as the fixed mathematical order.

Lemma 4.2 (Within-pair reindexing invariance). *Fix $\eta = (\eta_1, \dots, \eta_B) \in \{0, 1\}^B$, and reindex the two outcomes in block b by*

$$q_{bj}^\eta = q_{b,j \oplus \eta_b}, \quad j \in \{0, 1\},$$

where $\oplus$ denotes addition modulo two. If T^η is the statistic computed from the reindexed outcomes, then

$$T^\eta(z) = T(z \oplus \eta) \quad \text{for every } z \in \Omega.$$

Consequently, the complete paired-swap p -value is unchanged. The bounded-odds envelope in equation (17) is also unchanged.

Proof. Under reindexing, the condition-zero and condition-one samples at assignment z are exactly the original samples at assignment $z \oplus \eta$, respectively. This proves the displayed identity. The map $z \mapsto z \oplus \eta$ is a bijection of Ω , and it maps the reindexed observed assignment to the original observed assignment; hence it preserves both the observed statistic and its complete upper-tail count. For the sensitivity envelope, replace p_b by $1 - p_b$ whenever $\eta_b = 1$. This is a bijection of the symmetric probability box $[1/(1 + \Lambda), \Lambda/(1 + \Lambda)]^B$ and preserves corresponding assignment probabilities. Taking the maximum therefore gives the same envelope. $\square$

For quotient laws P and P' , independent $U, U' \sim P$, and independent $V, V' \sim P'$, the population discrepancy is

$$\text{MMD}_k^2(P, P') = \mathbb{E}k(U, U') + \mathbb{E}k(V, V') - 2\mathbb{E}k(U, V). \quad (13)$$

With the kernel in (9), $\text{MMD}_k(P, P') = 0$ if and only if $P = P'$, by proposition 3.2. The empirical statistic follows Gretton et al. [7]. For a positive-definite kernel k , we use the conventional two-sample formula

$$\begin{aligned} \widehat{\text{MMD}}_u^2(z) &= \frac{1}{B(B-1)} \sum_{\substack{b, b'=1 \\ b \neq b'}}^B k(q_{b,1-z_b}, q_{b',1-z_{b'}}) \\ &\quad + \frac{1}{B(B-1)} \sum_{\substack{b, b'=1 \\ b \neq b'}}^B k(q_{b,z_b}, q_{b',z_{b'}}) \\ &\quad - \frac{2}{B^2} \sum_{b, b'=1}^B k(q_{b,z_b}, q_{b',1-z_{b'}}). \end{aligned} \quad (14)$$

When the two samples are independent and identically distributed within arm, (14) is unbiased for (13). With matched blocks, within-pair dependence can remove that unbiasedness; here $\widehat{\text{MMD}}_u^2$ is a prespecified discrepancy statistic. The exactness of theorem 4.3 does not rely on unbiased estimation of a superpopulation parameter. For the primary endpoint, k is (9), with bandwidth $\hat{\sigma}$ determined by

$$\hat{\sigma}^2 = \frac{1}{2} \text{median} \left\{ \|z_i - z_j\|_2^2 : i < j, \|z_i - z_j\|_2^2 > 0 \right\},$$

where $z_i = \psi\{C(w_i)\}$ ranges over the pooled representations from the contrast being tested. If there is no positive pairwise distance, $\hat{\sigma} = 1$. This pooled, label-free rule is unchanged by treatment-label swaps.

4.2 Exact paired randomization test

For any deterministic statistic $T(z)$, including (14), define

$$p(Z) = 2^{-B} \sum_{z \in \Omega} \mathbf{1}\{T(z) \geq T(Z)\}. \quad (15)$$

This is the complete Fisher randomization distribution, not a Monte Carlo approximation [6, 13]. The floating-point implementation counts $T(z) \geq T(Z) - 10^{-12}$; this can only add upper-tail assignments and is therefore conservative relative to (15).

Theorem 4.3 (Finite-sample randomization validity). *Suppose assumption 4.1 holds, block retention is swap-invariant, and, conditional on the unordered eligible wells, their complete schedule of potential outcomes, and all pretreatment design information, Z is uniform on Ω . Under the Fisher sharp null that the invariant potential outcome of every confirmation well is unchanged by the two selected conditions,*

$$H_0^{\text{sharp}} : Q_{bj}(1) = Q_{bj}(0) \quad \text{for all } b \in \{1, \dots, B\}, j \in \{0, 1\},$$

the test that rejects when $p(Z) \leq \alpha$ satisfies

$$\mathbb{P}\{p(Z) \leq \alpha \mid \text{eligible wells, potential outcomes, and design information}\} \leq \alpha$$

for every $\alpha \in [0, 1]$.

Proof. Under the sharp null, all q_{bj} , every label-invariant tuning parameter, and therefore the collection $\{T(z) : z \in \Omega\}$ are fixed across the randomization distribution. Write $N = |\Omega| = 2^B$ and, for $z \in \Omega$,

$$r(z) = \#\{s \in \Omega : T(s) \geq T(z)\}.$$

Then $p(z) = r(z)/N$. For any integer m , at most m assignments can satisfy $r(z) \leq m$: after ordering the N values from largest to smallest, the upper-tail rank of the $(m+1)$ st value is at least $m+1$, and ties can only increase that rank. Taking $m = \lfloor \alpha N \rfloor$ gives

$$\#\{z : p(z) \leq \alpha\} \leq \lfloor \alpha N \rfloor \leq \alpha N.$$

Uniformity of Z on Ω proves the claimed conditional bound. $\square$

The public release does not contain the realized probability law of the original within-plate assignment algorithm. We therefore complement the uniform-assignment analysis by an exact bounded-odds sensitivity envelope in the spirit of matched-pair sensitivity analysis [18]. This analysis concerns possible departure from the reported randomization; it is not needed when uniform randomization is known from the design record.

Proposition 4.4 (Exact bounded-odds sensitivity envelope). *Let $\mathcal{F}$ contain the eligible wells, their complete potential-outcome schedule, and the pretreatment design information. For $\Lambda \geq 1$, suppose that, conditional on $\mathcal{F}$, the coordinates of Z are independent and*

$$\frac{1}{1+\Lambda} \leq p_b \leq \frac{\Lambda}{1+\Lambda}, \quad \mathbb{P}(Z = z \mid \mathcal{F}) = \prod_{b=1}^B p_b^{z_b} (1-p_b)^{1-z_b}. \quad (16)$$

For an observed assignment Z_{obs} , define

$$\bar{p}_\Lambda = \max_{p_b \in [1/(1+\Lambda), \Lambda/(1+\Lambda)]} \sum_{z \in \Omega} \mathbf{1}\{T(z) \geq T(Z_{\text{obs}})\} \prod_{b=1}^B p_b^{z_b} (1-p_b)^{1-z_b}. \quad (17)$$

The maximum in (17) is attained with every p_b at an endpoint of its interval. Consequently it is computed exactly by enumerating 2^B assignments and 2^B endpoint vectors. Under the Fisher sharp null, $\bar{p}_\Lambda$ is a valid worst-case upper-tail p -value for every assignment law satisfying (16); at $\Lambda = 1$, it equals (15).

Proof. For fixed p_k , $k \neq b$, the objective in (17) has the form $p_b A + (1 - p_b) B'$ for constants $A, B' \geq 0$. It is affine in p_b , so moving p_b to one of its two endpoints cannot decrease the objective. Repeating this coordinate by coordinate produces a maximizing vertex of the probability box, proving the enumeration claim.

For any fixed admissible vector $p = (p_1, \dots, p_B)$, the usual upper-tail probability

$$p_p(Z) = \mathbb{P}_p\{T(Z') \geq T(Z) \mid \mathcal{F}\}, \quad Z' \sim \mathbb{P}_p,$$

is super-uniform under the sharp null, including in the presence of ties. To see this directly, order the distinct statistic values as $t_1 > \dots > t_m$, put $\mu_r = \mathbb{P}_p\{T(Z) = t_r \mid \mathcal{F}\}$, and let $s_r = \sum_{j=1}^r \mu_j$. On the event $T(Z) = t_r$, one has $p_p(Z) = s_r$. Therefore, for any α , the probability of $p_p(Z) \leq \alpha$ is either zero or s_{r^*} , where $r^* = \max\{r : s_r \leq \alpha\}$; in either case it is at most α . Because $\bar{p}_\Lambda(Z) \geq p_p(Z)$ for the true admissible p , the event $\{\bar{p}_\Lambda(Z) \leq \alpha\}$ is contained in $\{p_p(Z) \leq \alpha\}$, establishing validity. If $\Lambda = 1$, every $p_b = 1/2$, and (17) reduces to the uniform average in (15). $\square$

The theorem is conditional and therefore permits arbitrary block effects and arbitrary dependence of the acquisition transformation on treatment and the intrinsic image: after maximal-invariant coding, that transformation does not alter $Q_{bj}(a)$. Exactness also requires that outcome-dependent selection not be performed in the confirmation blocks. We therefore select the condition pairs using discovery rows only, freeze the resulting manifest and record its SHA-256 digest, and compute no confirmation image feature or statistic before the selection and assignment audits are complete.

The primary contrast uses one exact test for the canonical-pixel endpoint. The nine other condition pairs form a prespecified secondary-contrast family; they are not biological replications of the primary contrast. Their exact p -values use Holm's step-down adjustment [8]. To summarize Monte Carlo rejection probabilities we report exact Clopper–Pearson intervals [2]. Paired-block percentile bootstrap quantiles describe the resampled statistic only; they are not confidence intervals and no coverage statement is attached to them.

5 Simulation study

5.1 Design

The simulation is a controlled test of causal inference when acquisition motions are hidden and informative. Each replicate contains 8 randomized blocks with two fixed units per block and one unit assigned to each treatment. A unit's intrinsic outcome has two sites, six channels, and a 28×28 lattice. A shared disk-and-satellite morphology induces block dependence, while site-specific location, radius, and intensity noise induce unit variation. The array is padded by eight pixels on every side before acquisition contamination.

Let $\eta \in \{0, 0.35, 0.70, 1.00\}$ be the intrinsic effect strength. Treatment adds a concentric annulus of amplitude 125η in the nuclear channel, 81.25η in the second channel, and 43.75η in the fifth channel. At $\eta = 0$, each unit's two potential outcomes are bitwise identical, so the Fisher sharp null in theorem 4.3 holds. After assignment, each site receives an integer shift of at most five pixels per coordinate and a quarter turn. The deterministic generator depends on treatment, a realized

image morphology score, unit identifier, and frozen replicate seed. Padding prevents loss of nonzero support. The analyzer receives neither the generated group elements nor the generator inputs.

Four procedures use the same Gaussian-kernel statistic and all $256 = 2^8$ paired assignments: transformed raw pixels; estimated moment registration; exact quotient pixels; and the finite Euler vector in (11). Moment registration translates the estimated intensity centroid to the canvas center, aligns the leading intensity-covariance eigenvector using interpolation, and resolves its π -ambiguity lexicographically. It is a practical registration comparator, not an exact invariant. The frozen experiment uses 250 independent replicates at each of the four effect strengths. Rejection fractions at level 0.05, Monte Carlo standard errors, and exact 95% Clopper–Pearson intervals are computed at every strength; the null interval evaluates type-I calibration.

5.2 Results

Table 1: Simulation paired-swap rejection fractions under treatment- and outcome-dependent acquisition transformations whose realized parameters are excluded from the estimators. The parenthesized interval at $\eta = 0$ is the exact 95% Clopper–Pearson interval across independent simulation replicates.

Method	$\eta = 0$	$\eta = 0.35$	$\eta = 0.70$	$\eta = 1.00$	Permutations
Exact quotient	0.0520 (0.0280, 0.0873)	0.1400	0.8480	0.9920	256
Raw pixels	1.0000 (0.9854, 1.0000)	1.0000	1.0000	1.0000	256
Moment registration	0.0560 (0.0310, 0.0922)	0.5400	0.9960	1.0000	256
Finite Euler signature	0.0400 (0.0193, 0.0723)	0.1840	1.0000	1.0000	256

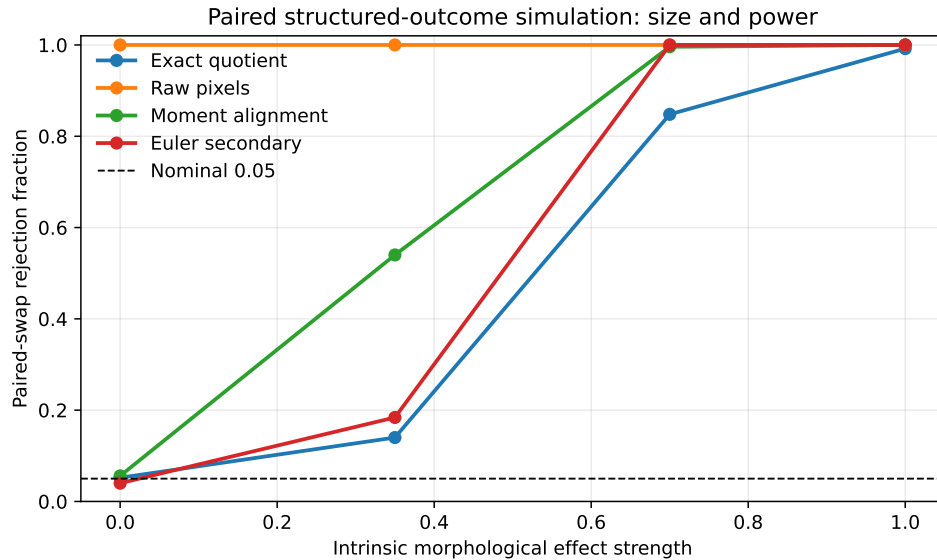


Figure 1: Paired-swap rejection fractions across intrinsic effect strengths. Finite-sample causal validity under informative acquisition pertains to the exact quotient and post-canonical Euler endpoints; raw coordinates and moment registration are included as diagnostics.

The null column in table 1 is a Monte Carlo evaluation of the complete randomization pipeline. For the exact quotient and post-canonical Euler tests, the positive- η columns estimate power against

a known intrinsic morphological effect. For the raw-coordinate and moment-registration diagnostics they are rejection fractions only, because those procedures do not have the invariant-endpoint validity guarantee of theorem 4.3 under informative acquisition.

6 RxRx1 confirmation experiment

6.1 Experimental units and frozen selection

The public RxRx1 release contains 125,510 six-channel 512×512 fluorescence microscopy images, with two nonoverlapping sites per well, 51 experimental batches, four cell types, and 1,138 siRNAs including 1,108 noncontrol perturbations [16, 24]. HUVEC contributes 24 batches. The experimental unit in our analysis is the well; sites and pixels are never treated as independent replicates. The causal conditions are the applied siRNA reagents, so the estimand is a reagent-condition contrast rather than a gene-specific biological effect.

RxRx1 partitions the 1,108 treatment siRNAs into four groups of 277 that remain together on a plate, and its documentation reports that their well locations were randomized within experiment and plate [24]. A paired randomization analysis must therefore compare conditions in the same group. We use HUVEC-01 through HUVEC-12 as 12 discovery batches, reserve HUVEC-13 through HUVEC-16, and use the image rows from HUVEC-17 through HUVEC-24 as 8 confirmation batches.

Selection uses the public 128-dimensional RxRx1 embeddings in discovery only. The two site embeddings are averaged within well, and each coordinate is median-centered and scaled by 1.4826 times its median absolute deviation over all noncontrol treatment wells within that discovery batch; any resulting scale below 10^{-6} is replaced by 1. Write $E_{je} \in \mathbb{R}^{128}$ for the normalized embedding of condition j in experiment e . Split the discovery batches into $H_1 = \{1, \dots, 6\}$ and $H_2 = \{7, \dots, 12\}$, and define

$$\begin{aligned} \bar{E}_{jh} &= |H_h|^{-1} \sum_{e \in H_h} E_{je}, & V_{jh} &= |H_h|^{-1} \sum_{e \in H_h} \|E_{je} - \bar{E}_{jh}\|_2^2, \\ \delta_h(j, k) &= \bar{E}_{jh} - \bar{E}_{kh}, & R_h(j, k) &= \frac{\|\delta_h(j, k)\|_2^2}{V_{jh} + V_{kh} + 10^{-12}}. \end{aligned}$$

For a pair (j, k) , its stability score is

$$S(j, k) = \min_{h \in \{1, 2\}} R_h(j, k) \max \left\{ \frac{\langle \delta_1(j, k), \delta_2(j, k) \rangle}{\sqrt{(\|\delta_1(j, k)\|_2^2 + 10^{-12})(\|\delta_2(j, k)\|_2^2 + 10^{-12})}}, 0 \right\}. \quad (18)$$

Eligibility requires exactly one two-site well for each condition in every discovery batch and an identical discovery-batch plate signature for the two conditions. Candidates are ordered by $S(j, k)$, then by descending minimum half-SNR, direction cosine, and mean half-SNR, followed by ascending integer identifiers; greedy matching gives 10 disjoint pairs. The S -highest pair is primary and the remaining 9 form a secondary-contrast panel. Neither confirmation metadata, confirmation embeddings, nor confirmation images enter this discovery-only selector.

The public release documents which rows are discovery/train and confirmation/test, but does not establish that the released embedding generator itself was trained independently of all confirmation images. Accordingly, the split certifies row-level exclusion; the assignment-based interpretation additionally assumes that the frozen condition list is independent of confirmation assignment outcomes.

The discovery-only selection manifest and its SHA-256 integrity digest are frozen before confirmation image analysis. A prior implementation also filtered on confirmation completeness; a replay that

removed every confirmation field returned exactly the same ten ordered pairs and scores. The earlier manifest is retained as an invalidated audit record. Before analysis, a separate confirmation metadata pass must establish completeness and same-plate occurrence; failure of either gate terminates the analysis. The checksum records integrity but is not an externally timestamped preregistration. Each eligible contrast has 8 paired confirmation blocks and therefore 256 assignments in its exact randomization distribution. The frozen confirmation subset contains 160 wells, 320 sites, and 1,920 single-channel PNG files.

6.2 Estimator-unobserved acquisition stress test

Each confirmation site is loaded as six 8-bit channels and embedded as a finite-support lattice array by 512-to-64 Pillow BOX resampling, retaining 8-bit values, and then adding a 16-pixel zero border. The nonzero support bounding box is therefore in $\mathcal{X}_{6,64}$, while the larger 96×96 storage canvas prevents cropping during the imposed motion.

For the six-channel array I_{ws} from well w and site s , define the coordinate-free morphology score and its pooled median by

$$m_{ws} = \text{mean}(I_{ws}/255) + 0.25 \text{ sd}(I_{ws}/255), \quad \tilde{m} = \text{median}_{w,s} m_{ws}.$$

For each published nuisance seed, the UTF-8 string `seed|well_id|site=s` is hashed to a 24-byte BLAKE2b digest, which is split consecutively into three little-endian unsigned 64-bit integers $H_{0,ws}, H_{1,ws}, H_{2,ws}$. With treatment label $a_w \in \{0, 1\}$, put

$$\begin{aligned} b_{ws} &= \mathbf{1}\{m_{ws} > \tilde{m}\}, & q_{ws} &= \lfloor 1000003 |m_{ws}| \rfloor, \\ r_{ws} &= (H_{0,ws} \bmod 2 + 2a_w + b_{ws}) \bmod 4, & d_w &= 2a_w - 1, \\ \Delta y_{ws} &= d_w \{1 + (H_{1,ws} + q_{ws}) \bmod 8\}, & \Delta x_{ws} &= d_w \{1 + (H_{2,ws} + 7q_{ws} + b_{ws}) \bmod 8\}. \end{aligned}$$

All six channels at a site receive the same quarter turn r_{ws} and translation $(\Delta x_{ws}, \Delta y_{ws})$, while the two sites receive separate, site-specific transformations. Thus the nuisance distribution is both treatment- and outcome-dependent and every realized transformation belongs to the declared product group. The realized transformation parameters are logged separately but are not supplied to any representation or test routine. No interpolation is used, and the software aborts if nonzero support is cropped.

This is a controlled unknown-acquisition experiment on real biological images: the content and batch structure are those of RxRx1, while the transformation law is imposed so that nuisance ground truth is available for audit but absent from estimator inputs. The claim evaluated is invariance to the declared lattice rigid-motion action, including when the nuisance is treatment- and outcome-dependent.

6.3 Endpoints, controls, and inference

The confirmatory endpoint is the two-site canonical pixel code (8), compared by Gaussian-kernel MMD and the complete 2^8 paired randomization distribution. The primary pair is tested at level 0.05. The same canonical-pixel statistic is then applied to the 9 prespecified secondary contrasts, whose exact p -values are adjusted together by Holm's method to control familywise error at 0.05. For the primary pair, the finite Euler signature is a secondary invariant summary. Raw transformed pixels and moment registration are acquisition-sensitive diagnostics; their paired-swap p -values are fully enumerated but do not inherit causal validity under informative acquisition. The prespecified analysis also generated 10000 paired-block bootstrap resamples. Resampling eight blocks with

replacement duplicates observations; applying the conventional observation-level U-statistic to those duplicated rows changes its diagonal-exclusion structure. The resulting quantiles are therefore retained as a descriptive audit of the resampled statistic, not as confidence intervals or precision bounds. Robustness to the transformation generator is evaluated over 20 frozen seeds. Seed 20260731 is the prespecified primary acquisition realization used for the primary table, its four bootstrap-statistic quantile pairs, the secondary-contrast panel, and the approximate-action baseline; the other 19 seeds assess robustness of exact invariance and acquisition-only behavior. Software tests and internal orbit audits are completed before the outcome tables are generated.

Two controls separate exact group invariance from approximate robustness. First, for every pair and confirmation block, an acquisition-only sharp-null prototype is constructed by averaging the two clean well arrays and rounding back to 8-bit values; the same prototype is assigned to both conditions before the label-dependent hidden motion is applied. This guarantees equality of the intrinsic potential outcomes while retaining informative acquisition. The resulting rejection fractions are descriptive stress-test summaries over pairs and nuisance seeds; the independent simulation provides the formal Monte Carlo type-I assessment. Second, the primary pair is perturbed outside the declared action by interpolated rotations of 5° , 10° , and 20° , two-pixel support truncation, and within-support Gaussian noise of standard deviation two. Changes in features, test statistics, and p -values quantify sensitivity when the acquisition process only approximately follows the action. Before these perturbations, every site in the seed-20260731 contaminated primary-pair arrays is support-centered. For method m , if $Z_m^{(0)}$ and $Z_m^{(\delta)}$ are the resulting baseline and perturbed feature matrices, the reported relative error is

$$\varepsilon_m(\delta) = \frac{\|Z_m^{(\delta)} - Z_m^{(0)}\|_F}{\max\{\|Z_m^{(0)}\|_F, 10^{-12}\}}.$$

The label-free median bandwidth rule is recomputed from the pooled primary-pair features for every perturbation before its statistic and complete paired-swap p -value are evaluated.

6.4 Results

The primary contrast is siRNA 384 (**s38230**) versus siRNA 747 (**s37149**). The confirmation completeness and same-plate gate passed for all frozen pairs and blocks. The discovery-only selection manifest has SHA-256 **b6f350ca96fb9e7a6271845dd6e432ed9dca556144a1a332d8f5af168642b968**.

Table 2: RxRx1 analysis for the primary pair under imposed lattice rigid motions whose parameters are excluded from estimator inputs. The displayed kernel discrepancy is defined in (14), and p -values enumerate all paired label swaps. The canonical-pixel endpoint is the sole primary test. The Euler p -value is an unadjusted secondary analysis; raw and moment-registration values are diagnostics. The finite-sample causal validity theorem applies to the invariant quotient and post-canonical Euler rows, not to the raw or moment-registration diagnostics.

Representation	$\widehat{\text{MMD}}_u^2$	Paired-swap p
Canonical quotient pixels	0.0036	0.0078
Finite directional Euler signature	0.7535	0.0078
Moment registration	-0.0065	0.2109
Raw transformed pixels	0.1664	0.0078

0 of the nine frozen secondary contrasts remained significant after Holm adjustment.

Table 3: Prespecified paired-block bootstrap audit for the primary contrast. The last column contains the empirical 2.5th and 97.5th percentiles of the resampled U-statistic over 10000 draws. Because duplicate blocks alter the conventional observation-level U-statistic’s diagonal-exclusion structure at $B = 8$, these quantiles are not confidence intervals and are not interpreted as uncertainty bounds around the point statistic.

Representation	Original $\widehat{\text{MMD}}_u^2$	Bootstrap-statistic quantiles
Canonical quotient pixels	0.0036	(0.0415, 0.3505)
Finite directional Euler signature	0.7535	(0.5582, 1.2290)
Moment registration	-0.0065	(0.0297, 0.3431)
Raw transformed pixels	0.1664	(0.2008, 0.5021)

Table 4: Canonical-quotient tests for the prespecified secondary-contrast panel. Holm adjustment is applied to the family of 9 secondary-contrast p -values; these are different siRNA contrasts, not biological replications of the primary contrast.

Pair	Conditions	$\widehat{\text{MMD}}_u^2$	Swap p	Holm p
1	s21474 versus s21486	0.0040	0.0859	0.1719
2	s29342 versus s28343	0.0045	0.0156	0.0703
3	s18592 versus s18763	0.1266	0.0078	0.0703
4	s20782 versus s21523	0.0031	0.0078	0.0703
5	s20401 versus s19178	0.0228	0.0156	0.0703
6	s29371 versus s28116	0.0063	0.0078	0.0703
7	s17831 versus s20468	0.0486	0.0078	0.0703
8	s37483 versus s224918	-0.0107	0.1562	0.1719
9	s37712 versus s224940	0.0761	0.0078	0.0703

Across all acquisition-transform seeds, the maximum absolute clean-versus- contaminated feature discrepancies were 0.0000 for the maximal-invariant pixels and 0.0000 for the finite Euler signature. For the maximal-invariant endpoint, the largest corresponding changes in $\widehat{\text{MMD}}_u^2$ and its exact p -value were 0.0000 and 0.0000, respectively.

Table 5: Acquisition-only sharp-null stress test over frozen pairs and hidden nuisance seeds. The intrinsic array is identical for both conditions within each block before the treatment-dependent acquisition motion.

Method	Tests	Rejections at 0.05	Rejection fraction
Exact quotient	200	0	0.0000 (0.0000, 0.0183)
Finite Euler signature	200	0	0.0000 (0.0000, 0.0183)
Moment registration	200	0	0.0000 (0.0000, 0.0183)
Raw pixels	200	200	1.0000 (0.9817, 1.0000)

Parentheses give exact binomial Clopper–Pearson intervals as descriptive summaries only; pair–seed tests share biological blocks and are not independent Bernoulli trials.

Table 6: Approximate-action stress test for the primary pair. Feature error is relative to the exactly transformed baseline; the last two columns give absolute changes from the baseline statistic and exact p -value.

Perturbation	Method	Relative feature error	$ \Delta \widehat{\text{MMD}}_u^2 $	$ \Delta p $
Rotation 5°	Exact quotient	0.8432	0.0012	0.0000
Rotation 5°	Finite Euler signature	0.0779	0.0860	0.0000
Rotation 5°	Moment registration	0.7295	0.0014	0.0234
Rotation 5°	Raw pixels	0.6540	0.0013	0.0000
Rotation 10°	Exact quotient	0.8967	0.0022	0.0000
Rotation 10°	Finite Euler signature	0.0827	0.0530	0.0000
Rotation 10°	Moment registration	0.8124	0.0021	0.1250
Rotation 10°	Raw pixels	0.8038	0.0012	0.0000
Rotation 20°	Exact quotient	0.9715	0.0005	0.0000
Rotation 20°	Finite Euler signature	0.1076	0.0898	0.0000
Rotation 20°	Moment registration	0.8662	0.0009	0.0469
Rotation 20°	Raw pixels	0.8802	0.0014	0.0000
Two-pixel support crop	Exact quotient	0.8253	0.0026	0.0000
Two-pixel support crop	Finite Euler signature	0.1478	0.0164	0.0000
Two-pixel support crop	Moment registration	0.6857	0.0203	0.2031
Two-pixel support crop	Raw pixels	0.3523	0.0001	0.0000
Within-support Gaussian noise	Exact quotient	0.5949	0.0007	0.0000
Within-support Gaussian noise	Finite Euler signature	0.0618	0.0244	0.0000
Within-support Gaussian noise	Moment registration	0.3870	0.0067	0.1328
Within-support Gaussian noise	Raw pixels	0.1775	0.0006	0.0000

6.5 Post-analysis sensitivity and persistence diagnostics

The following two diagnostics were specified after the confirmatory outcomes and primary result had been accessed. They neither alter the frozen endpoint nor convert the extension into a

confirmatory analysis. First, we evaluated proposition 4.4 for the primary canonical-quotient statistic. The calculation enumerated all 256 assignments and all 256 vertices of the bounded-odds probability box at each value of Λ . The machine-readable extension audit has SHA-256 7928f075dd02630e2c735a89bbd64e8a76f8b9191b2785cfe8de6542462c0882.

Table 7: Post-analysis sensitivity of the primary quotient test to departure from uniform paired assignment. The second column is the exact worst-case upper-tail probability under the independent product model in (16).

Sensitivity factor Λ	Worst-case upper p
1.00	0.007812
1.10	0.008311
1.25	0.010597
1.50	0.017452
2.00	0.039171
3.00	0.100128
5.00	0.232569

At $\Lambda = 1$, the envelope reproduces the complete paired-swap value 0.0078125. The worst-case upper probability first reaches 0.05 at $\Lambda = 2.199995$ (numerical upper probability 0.050000). Thus, within the conditional-independence model (16), the primary conclusion remains below 0.05 for smaller sensitivity factors. This statement quantifies possible nonuniformity of the reported assignment mechanism; it is not an estimate of a specific unmeasured confounder’s effect.

Second, we evaluated a lower-star, zero-dimensional persistence diagnostic on the 16-by-16 resampled invariant nuclear-channel representation for all 32 primary-pair sites. All 32 diagrams were exactly invariant under the recorded hidden lattice actions, with maximum bottleneck distance 0.000000. After an independently generated one-gray-level bounded perturbation, all 32 diagram comparisons satisfied the bottleneck stability inequality and all 32 diagnostic-Gaussian-metric inversion checks passed. The largest bottleneck distance was 0.003918, no larger than the largest realized sup-norm perturbation 0.003920. This audit concerns the stable grayscale filtration in theorem 3.3, not the thresholded finite Euler signature.

The internal algebraic and software audit recorded 0 canonical-code failures among 6400 generated transformations. The software test suite passed 45 checks. Its source fingerprint is `source-e9f042c387cf`.

7 Discussion

The observation model (1) permits a particularly adversarial nuisance: acquisition may depend on both treatment and outcome. Under that model, theorem 2.1 gives a sharp boundary. An invariant target is observed exactly; a noninvariant target cannot be recovered uniformly without additional information. A maximal invariant reaches this boundary by retaining the full orbit. Causal identification remains a separate step, supplied here either by quotient-specific exchangeability in theorem 2.11 or by the randomization design under the conditional uniformity premise motivated by the reported within-plate randomization in theorem 4.3.

The lattice construction also shows that compactness is not necessary when an explicit measurable cross-section is available. Integer translations form a noncompact group, yet support normalization followed by finite rotational canonicalization is exact and maximal. This avoids a

nuisance-distribution model and preserves every feature that is identifiable modulo the declared rigid motions. The Gaussian quotient kernel is characteristic for quotient laws and supplies the discrepancy statistic used in the sharp-null test, whereas the finite Euler vector answers a deliberately lower-dimensional secondary question.

The padding and support checks in the controlled experiment ensure that the imposed translations preserve the full nonzero support. If a natural field of view clips that support, the appropriate model is $X = \Pi\{\Gamma \cdot Y(A)\}$, and theorem 2.3 replaces ordinary orbit invariance. In particular, if two distinct intrinsic orbits can produce the same crop, their maximal-orbit label is not recoverable from that crop. The support-truncation stressor removes a margin from the realized support bounding box; it demonstrates sensitivity to loss of intrinsic content but is not a simulation of translation through a fixed sensor boundary and does not restore identification.

Abstract continuous rotations are covered by corollary 2.8; the unresolved computational issue is an exact, useful representation for the chosen continuous action. On a pixel lattice, interpolated rotation is not the same operation as the exact C_4 permutation used here. Similarly, corollary 2.12 identifies conditional effects only after an integrable scalar invariant functional and adequate covariates are specified. With eight primary blocks and no prespecified subgroup model, fitting a high-dimensional heterogeneous-effect learner would not support credible inference, so none is reported.

The scientific interpretation is action-specific. The empirical intervention validates the method against imposed integer translations and quarter turns whose realized elements are excluded from estimator inputs; it does not convert general optical deformation, rescaling, or changing cell content into members of that action. Likewise, inference from the reported RxRx1 within-plate randomization concerns the selected siRNA reagents in the analyzed HUVEC plate design. These qualifications delimit the estimand rather than weaken identification: within the stated action, invariance is algebraic; under the conditional uniformity, no-interference, well-defined-version, and swap-invariant-retention assumptions, randomization validity is finite-sample exact for the invariant Fisher sharp null.

8 Conclusion

Unknown unit-specific transformations need not be estimated in order to conduct causal inference on the information that survives them. For measurable group contamination, orbit invariance is necessary and sufficient for exact observability, and a Borel maximal invariant carries every such target. We provide a complete identification formula for its interventional law, an explicit maximal invariant for multichannel lattice images under $\mathbb{Z}^2 \rtimes C_4$, and an exact paired randomization test using a characteristic quotient kernel. The simulation and RxRx1 discovery–confirmation experiments turn these population statements into an auditable workflow with informative acquisition transformations whose realized parameters are excluded from estimator inputs.

Reproducibility statement

The accompanying software downloads the frozen RxRx1 subset, verifies file and manifest hashes, displays progress for data acquisition and every simulation stage, constructs all four representations, enumerates the complete assignment distribution, computes bootstrap and multiplicity-adjusted summaries, executes the unit tests, and writes the numerical macros consumed by this manuscript. The analysis uses only public RxRx1 data subject to its stated license [16].

References

- [1] Satarupa Bhattacharjee, Bing Li, Xiao Wu, and Lingzhou Xue. Doubly robust estimation of causal effects for random object outcomes with continuous treatments. *arXiv preprint arXiv:2506.22754*, 2025. doi: 10.48550/arXiv.2506.22754.
- [2] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
- [3] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007. doi: 10.1007/s00454-006-1276-5.
- [4] Justin Curry, Sayan Mukherjee, and Katharine Turner. How many directions determine a shape and other sufficiency results for two topological transforms. *Transactions of the American Mathematical Society, Series B*, 9:1006–1043, 2022. doi: 10.1090/btran/122.
- [5] Morris L. Eaton. *Group Invariance Applications in Statistics*, volume 1 of *Regional Conference Series in Probability and Statistics*. Institute of Mathematical Statistics, Hayward, CA, 1989. doi: 10.1214/cbms/1462061021.
- [6] Ronald A. Fisher. *The Design of Experiments*. Oliver and Boyd, Edinburgh, 1935.
- [7] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012. URL <https://jmlr.org/papers/v13/gretton12a.html>.
- [8] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- [9] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.
- [10] Alexander S. Kechris. *Classical Descriptive Set Theory*, volume 156 of *Graduate Texts in Mathematics*. Springer, New York, 1995. doi: 10.1007/978-1-4612-4190-4.
- [11] Kwangho Kim and Hajin Lee. Topological causal effects. In *Proceedings of the Fourteenth International Conference on Learning Representations*, 2026. URL <https://iclr.cc/virtual/2026/poster/10008398>.
- [12] Daisuke Kurisu, Yidong Zhou, Taisuke Otsu, and Hans-Georg Müller. Geodesic causal inference. *arXiv preprint arXiv:2406.19604*, 2024. doi: 10.48550/arXiv.2406.19604.
- [13] E. L. Lehmann and Joseph P. Romano. *Testing Statistical Hypotheses*. Springer, New York, 3 edition, 2005. doi: 10.1007/0-387-27605-X.
- [14] Nobuyuki Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, 1979. doi: 10.1109/TSMC.1979.4310076.
- [15] Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and Wasiur R. KhudaBukhsh. Kernel-based estimators for functional causal effects. *arXiv preprint arXiv:2503.05024*, 2025. doi: 10.48550/arXiv.2503.05024.
- [16] Recursion. RxRx1: An image set for cellular morphological variation across many biological perturbations. <https://www.rxxr.ai/rxxr1>, 2023. Accessed 2026-07-31.

- [17] James M. Robins. A new approach to causal inference in mortality studies with a sustained exposure period—application to control of the healthy worker survivor effect. *Mathematical Modelling*, 7(9–12):1393–1512, 1986. doi: 10.1016/0270-0255(86)90088-6.
- [18] Paul R. Rosenbaum. Sensitivity analysis for certain permutation inferences in matched observational studies. *Biometrika*, 74(1):13–26, 1987. doi: 10.1093/biomet/74.1.13.
- [19] Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983. doi: 10.1093/biomet/70.1.41.
- [20] Donald B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974. doi: 10.1037/h0037350.
- [21] Amir Saki and Usef Faghihi. Beyond means: Topological causal effects under persistent-homology ignorability. *arXiv preprint arXiv:2603.14169*, 2026. doi: 10.48550/arXiv.2603.14169.
- [22] Ha-Young Shin, Kyusoon Kim, Kwonsang Lee, and Hee-Seok Oh. Absolute average and median treatment effects as causal estimands on metric spaces. *arXiv preprint arXiv:2407.03726*, 2024. doi: 10.48550/arXiv.2407.03726.
- [23] Bharath K. Sriperumbudur, Arthur Gretton, Kenji Fukumizu, Bernhard Schölkopf, and Gert R. G. Lanckriet. Hilbert space embeddings and metrics on probability measures. *Journal of Machine Learning Research*, 11:1517–1561, 2010. URL <https://jmlr.org/papers/v11/sriperumbudur10a.html>.
- [24] Maciej Sypetkowski, Morteza Rezanejad, Saber Saberian, Oren Kraus, John Urbanik, James Taylor, Ben Mabey, Mason Victors, Jason Yosinski, Alborz Rezazadeh Sereshkeh, Imran Haque, and Berton Earnshaw. RxRx1: A dataset for evaluating experimental batch correction methods. *arXiv preprint arXiv:2301.05768*, 2023. doi: 10.48550/arXiv.2301.05768.
- [25] Katharine Turner, Sayan Mukherjee, and Doug M. Boyer. Persistent homology transform for modeling shapes and surfaces. *Information and Inference: A Journal of the IMA*, 3(4):310–344, 2014. doi: 10.1093/imaiai/iau011.